Update on Project 7: Security

Cody Gunton - March 11, 2026 https://codygunton.github.io/talks-and-writing/2026-03-11-zkevm-breakout/

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RISC-V Compliance Testing

There are recent improvements to https://github.com/riscv/riscv-arch-test RISCOF framework, more invasive to implement, is no more.
New tests!

Commitment to more legible release schedule.

TODO: update https://eth-act.github.io/zkevm-test-monitor/

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Soundcalc

What it is: a Python tool to calculate number of bits of security.
Five zkVMs integrated by requested deadline https://github.com/ethereum/soundcalc/blob/main/reports/summary.md Thanks to the teams behind: Airbender, OpenVM, Pico, SP1, ZisK

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What does formal verification give us...?

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SNARK Specifications

https://codygunton.github.io/pil2-proofman/: Markdown guide to the Python implementation, the latter being the proposed starter.

ZisK Prover Specification
This is the complete specification for the PIL2 proving system as used by the ZisK zkVM. It specifies exactly what the prover
and verifier compute, in sequence, using mathematical notation. It contains no proofs and no security analysis—only the concrete protocol.

The specification comprises three parts:

1. STARK Protocol — the parametric FRI-STARK proving system that works with any AIR over the Goldilocks field.
2. ZisK Machine — the concrete chip architecture: 21 AIRs, bus interconnections, memory layout, coprocessors, precompile circuits, lookup tables, and global constraints.
3. Recursion Pipeline — the VADCOP recursive aggregation from per-AIR STARKs to a single proof.

This specification is accompanied by a Python executable specification that implements the complete STARK prover and verifier, producing byte-identical proofs to the production C++ implementation.

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Thanks for your attention!

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